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KONE Elevators

MATERIAALIN KÄYTTÖNOTTOSELVITYS

Numero (annetaan standardisoinnissa)

KOS ECO140

1. Osasto/Laattija/fax/tel HAT / A. Brusila / 2729 / 3318	2. PVM 25.10.94
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3. Nimitys Schottky diodi
4. Spesifikaatio $I_F (A_V) = \min 1.5 A, V_{FM} = 0.55 V, V_R = 40 V$
5. Lisäspesifikaatio

6. Valmistaja I International Rectifier	
7. Valmistajan tyyppi B5MQ040	
8. Toimittaja Flinkenberg, Ericsson Standard Components	*) Testaustapa B

9. Valmistaja II General Instrument	
10. Valmistajan tyyppi SS24	
11. Toimittaja Farnell Electronic Services	*) Testaustapa B

12. Valmistaja III Motorola	
13. Valmistajan tyyppi MBRS30LT3	
14. Toimittaja Arrow - Field	*) Testaustapa B

*) Testaustapa: a = valittu datakirjan perusteella, ei testattu b = toiminnallinen testi c = tyyppitesti d = elinikätesti
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15. HUOM!

16. Viitestandardi	17. Suositusluokka
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18. Ryhmäkoodi 740012	19. Identifikaatio 201 944
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HUOM! Laita AINA datakirjaviite tai kopiot valmistajien spesifikaatioista mukaan.

15MQ040



Voltage Ratings

Part number	15MQ040
V_R Max. DC Reverse Voltage (V)	40
V_{RWM} Max. Working Peak Reverse Voltage (V)	

Electrical Specifications

Parameters	15MQ	Units	Conditions
$I_{F(AV)}$ Max. Average Output Current	1.7	A	180° conduction, rectangular waveform
	1.5	A	180° conduction, sinusoidal waveform
I_{FSM} Max. Peak One Cycle Non-Repetitive Surge Current	60	A	50Hz half cycle sine wave or 6ms rectangular pulse
	64	A	60Hz half cycle sine wave or 5ms rectangular pulse
V_{FM} Max. Peak Forward Voltage	0.55	V	$T_J = 25^\circ\text{C}$ at rated $I_{F(AV)}$
I_R Max. dc Reverse Current	5.0	mA	$T_J = 25^\circ\text{C}$ $V_R = \text{rated } V_R$
I_{RM} Max. Peak Reverse Current	50	mA	$T_C = 125^\circ\text{C}$ $V_R = \text{rated } V_R$
C_T Typical Junction Capacitance	70	pF	$T_C = 25^\circ\text{C}$ $V_R = 10V_{DC}$ 1Mhz test signal

Thermal-Mechanical Specifications

Parameters	15MQ	Units	Conditions
T_J Max. Operating Junction Temperature	-40 to 125	$^\circ\text{C}$	
T_{stg} Max. Storage Temperature Range	-40 to 125	$^\circ\text{C}$	
$R_{\theta JA}$ Max. Thermal Resistance Junction to Ambient	150	$^\circ\text{C/W}$	
wt Approximate Weight	0.07(0.026)	g (oz.)	
Case Style	D-64		

SS22 THRU SS26

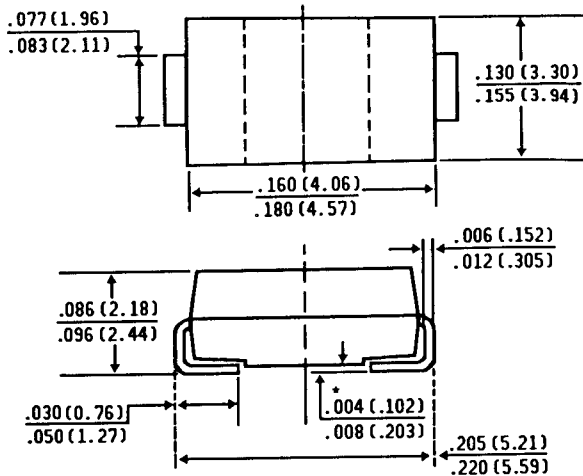
SURFACE MOUNT SCHOTTKY BARRIER RECTIFIER
VOLTAGE - 20 - 60 Volts CURRENT - 2.0 Amperes

FEATURES

- ◆ For surface mounted applications
- ◆ Low profile package
- ◆ Built-in strain relief
- ◆ Metal to silicon rectifier, majority carrier conduction
- ◆ Low power loss, high efficiency
- ◆ High current capability, low V_F
- ◆ High surge capacity
- ◆ Epitaxial construction
- ◆ For use in low voltage high frequency inverters, free wheeling, and polarity protection applications
- ◆ Guardring for transient protection



SMB/DO-214AA



Dimensions in inches and (millimeters)

*Typical Range

MECHANICAL DATA

Case: Molded plastic

Terminals: Solder plated

Polarity: Indicated by cathode band

Standard Packaging: 12mm tape
(EIA STD RS-481)

Weight: 0.003 ounces 0.093 gram

MAXIMUM RATINGS AND ELECTRICAL CHARACTERISTICS

Ratings at 25°C ambient temperature unless otherwise specified.
Resistive or inductive load.
For capacitive load, derate current by 20%.

	SYMBOLS	SS22	SS23	SS24	SS25	SS26	UNITS
Maximum Recurrent Peak Reverse Voltage	V _{RRM}	20	30	40	50	60	Volts
Maximum RMS voltage	V _{RMS}	14	21	28	35	42	Volts
Maximum DC Blocking Voltage	V _{DC}	20	30	40	50	60	Volts
Maximum Average Forward Rectified Current at T _L (See Figure 1)	I _(AV)	2.0					Amps
Peak forward Surge Current 8.3ms single half sine-wave superimposed on rated load (JEDEC Method)	I _{FSM}	50.0					Amps
Maximum Instantaneous Forward Voltage at 2.0A (NOTE 1)	V _F	0.50			0.70		Volts
Maximum DC Reverse Current (NOTE 1) T _A =25°C at Rated DC Blocking Voltage T _A =100°C	I _R	0.5					mA
		20.0			10.0		
Maximum Thermal resistance (NOTE 2)	R _{θJL}	20.0					°C/W
	R _{θJA}	100.0					
Operating Temperature Range	T _J	-65 to +125			-65 to +150		°C
Storage Temperature Range	T _{STG}	-65 to +150					°C

NOTES: 1. Pulse Test with $PW=300 \mu\text{sec}$, 2% Duty Cycle.

2. 5.0mm^2 (.013mm thick) land areas.

**GENERAL
INSTRUMENT**

MAXIMUM RATINGS AND CHARACTERISTIC CURVES SS22 THRU SS26

FIG. 1 - FORWARD DERATING CURVE

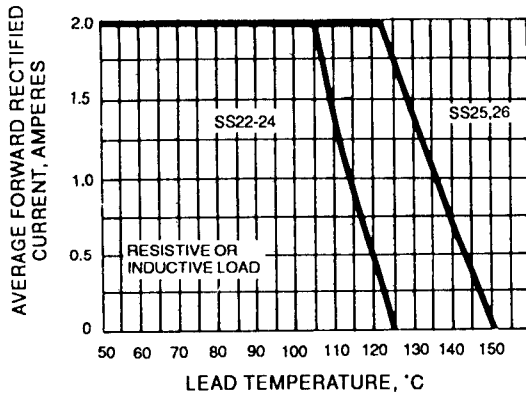


FIG.2- TYPICAL INSTANTANEOUS FORWARD CHARACTERISTICS

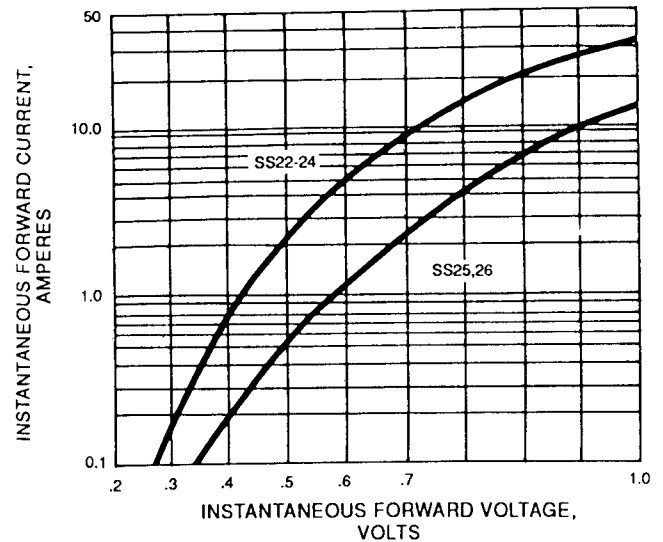


FIG.3 - TYPICAL REVERSE CHARACTERISTICS

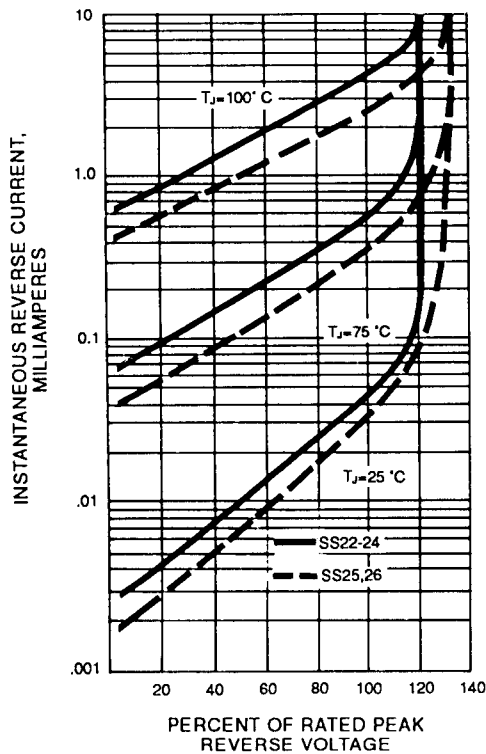


FIG.4 - TYPICAL JUNCTION CAPACITANCE

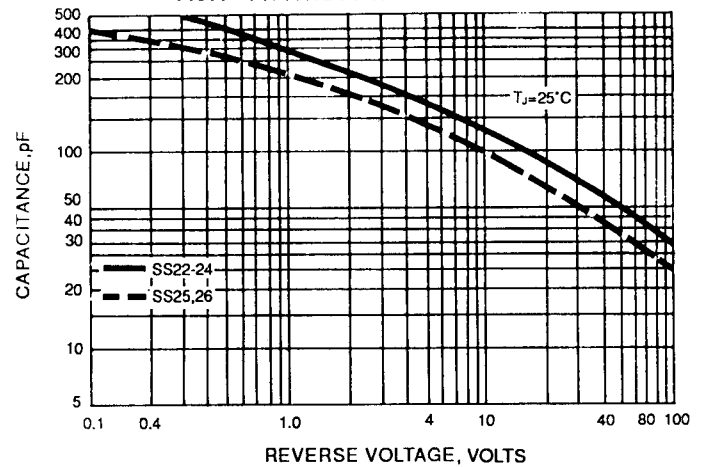
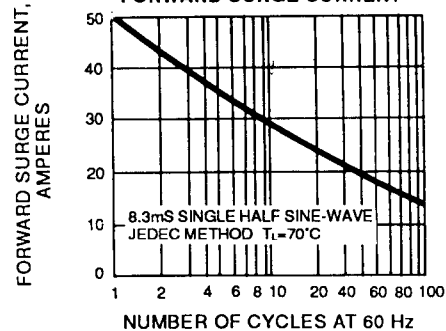


FIG 5 - MAXIMUM NON-REPETITIVE PEAK FORWARD SURGE CURRENT



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**GENERAL
INSTRUMENT**

**MOTOROLA
SEMICONDUCTOR
TECHNICAL DATA**

Designer's™ Data Sheet
**Schottky Power Rectifier
Surface Mount Power Package**

MBRS130LT3

Motorola Preferred Device

**SCHOTTKY BARRIER
RECTIFIER
1.0 AMPERE
30 VOLTS**



CASE 403A-03

Employs the use of the Schottky Barrier principle in a large area metal-to-silicon power diode. State-of-the-art geometry features epitaxial construction with oxide passivation and metal overlay contact. Ideally suited for low voltage, high frequency rectification, or as free wheeling and polarity protection diodes, in surface mount applications where compact size and weight are critical to the system.

- Very Low Forward Voltage Drop (0.395 Volts Max @ 1.0 A, $T_J = 25^\circ\text{C}$)
- Small Compact Surface Mountable Package with J-Bend Leads
- Packaged in 12 mm Pocket Tape and Reel
- Highly Stable Oxide Passivated Junction
- Guardring for Stress Protection

Mechanical Characteristics

Case: Transfer Molded Plastic Package, Case 403A-03

Lead Finish: Plated Leads, Readily solderable in surface Mount Applications

Polarity Identification: Notch in Plastic Body Indicates Cathode Lead

Device Marking: 1BL3

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	V_{RRM} V_{RWM} V_R	30	Volts
Average Rectified Forward Current $T_L = 120^\circ\text{C}$ $T_L = 110^\circ\text{C}$	$I_F(AV)$	1.0 2.0	Amps
Nonrepetitive Peak Surge Current (Surge applied at rated load conditions halfwave, single phase, 60 Hz)	I_{FSM}	40	Amps
Operating Junction Temperature	T_J	-65 to +125	$^\circ\text{C}$

THERMAL CHARACTERISTICS

Thermal Resistance — Junction to Lead ($T_L = 25^\circ\text{C}$)	$R_{\theta JL}$	12	$^\circ\text{C/W}$
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ELECTRICAL CHARACTERISTICS

Maximum Instantaneous Forward Voltage (1) ($I_F = 1.0\text{ A}$, $T_J = 25^\circ\text{C}$) ($I_F = 2.0\text{ A}$, $T_J = 25^\circ\text{C}$)	V_F	0.395 0.445	Volts
Maximum Instantaneous Reverse Current (1) (Rated dc Voltage, $T_J = 25^\circ\text{C}$) (Rated dc Voltage, $T_J = 100^\circ\text{C}$)	I_R	1.0 10	mA

(1) Pulse Test: Pulse Width = 300 μs , Duty Cycle = 2%

Preferred devices are Motorola recommended choices for future use and best overall value

Designer's Data for "Worst Case" Conditions — The Designer's Data Sheet permits the design of most circuits entirely from the information presented. Limit curves — representing boundaries on device characteristics — are given to facilitate "worst case" design.

MBRS130LT3

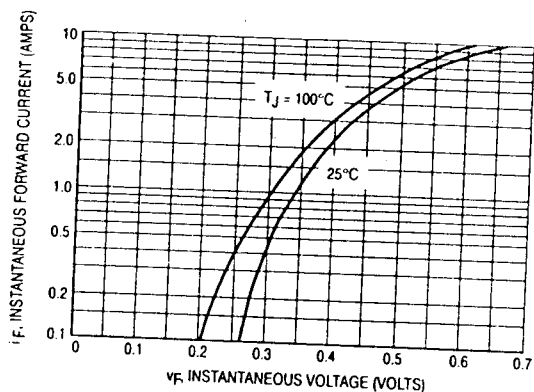


Figure 1. Typical Forward Voltage

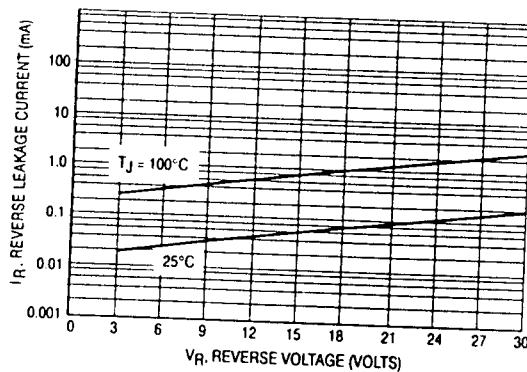


Figure 2. Typical Reverse Leakage Current

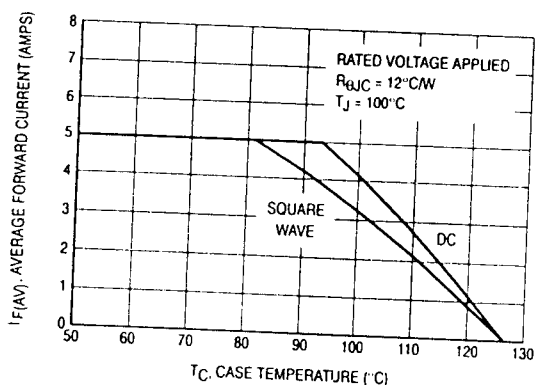


Figure 3. Current Derating (Case)

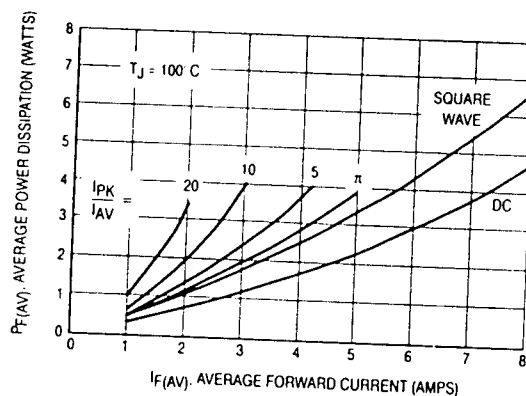


Figure 4. Power Dissipation

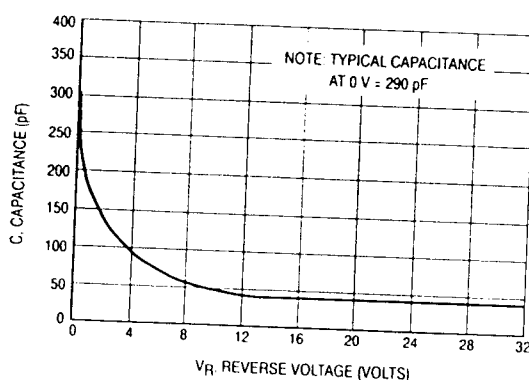


Figure 5. Typical Capacitance